

Project Plan for Global AI Hackathon Series with Qwen Cloud: Track 2 – AI Showrunner

- The project converts books, manga, or written stories in PDF format into short videos or film adaptations using Qwen models and video generation tools like Wan and HappyHorse.
- It features an Apple Books-style interface with a virtual bookshelf for browsing and selecting books, and a workspace with synchronized PDF and video views.
- The system includes two tabs: end-user mode for viewing videos and director mode for editing with pointer-based comments and targeted prompts.
- The backend employs multiple Qwen agents (Qwen 3.7 Max and Plus) collaborating to parse PDFs, extract narrative elements, scriptwrite, storyboard, and edit videos.
- The project aligns with Track 2's focus on autonomous short drama creation, narrative ability, multimodal orchestration, and maximizing output quality under token constraints.

Project Overview

The proposed project aims to create an AI-powered application that converts written stories in PDF format into engaging short videos or film adaptations. Leveraging Qwen Cloud's advanced AI models (Qwen 3.7 Max and Plus) and video generation tools (Wan, HappyHorse), the system autonomously handles the entire short drama creation pipeline—from scriptwriting and storyboarding to video generation and editing. The user interface mimics Apple Books, providing a virtual bookshelf for browsing and selecting books, with a workspace that displays the PDF alongside the generated video in synchronized scrolling. The interface includes two modes: end-user mode for viewing videos and director mode for editing, where users can add pointer-based comments and targeted prompts to refine the AI's output. This project addresses the growing demand for accessible, engaging visual storytelling, especially for users who prefer video over text.

Technical Architecture

Collaborative Multi-Agent System

The backend architecture employs multiple Qwen AI agents working in concert to decompose the complex task of converting PDF books into videos. The agents collaborate as follows:

- **PDF Parsing and Distillation Agent:** Extracts key narrative elements, context, and structural details from the uploaded PDF. This agent processes the text to identify characters, plot arcs, settings, and themes, distilling the content into a structured format



suitable for video script generation. It leverages Qwen 3.7 Max's 1M-token context window and advanced reasoning capabilities to handle long documents efficiently.

- **Scriptwriting Agent:** Takes the distilled narrative elements and generates a coherent script, ensuring fidelity to the source material while optimizing for video format constraints. This agent uses Qwen 3.7 Plus's multimodal capabilities to understand text and images, enhancing script quality.
- **Storyboarding and Scene Selection Agent:** Translates the script into a visual storyboard, selecting scenes and framing shots for video generation. This agent coordinates with the video generation tools (Wan, HappyHorse) to plan the visual narrative structure.
- **Video Generation and Editing Agent:** Uses Wan and HappyHorse APIs to generate video clips from the storyboard, ensuring cinematic quality, smooth motion, and narrative coherence. This agent also handles editing, including transitions, audio integration, and final assembly of the video.
- **Human-in-the-Loop (HITL) Checkpoints:** Implements HITL checkpoints at critical stages (e.g., script approval, storyboard review, video editing) using frameworks like LangGraph or Microsoft Agent Framework. These checkpoints allow human reviewers to pause the workflow, provide feedback, and approve or refine AI outputs, ensuring quality control and compliance with ethical guidelines.

Memory Management and Task Decomposition

- **Persistent Memory:** The system maintains a persistent memory of user preferences, project history, and feedback to improve future runs and personalize the experience. This memory is managed via vector storage and retrieval pipelines optimized for Alibaba Cloud infrastructure.
- **Task Decomposition:** The workflow is decomposed into discrete stages (PDF parsing, scriptwriting, storyboarding, video generation, editing) with clear handoffs between agents. Each agent operates within token limits, with fallback mechanisms to handle rate limits and errors gracefully.
- **Multimodal Orchestration:** The system integrates text, image, and video modalities, coordinating Qwen's text understanding with Wan and HappyHorse's video generation capabilities. This orchestration ensures narrative coherence and visual fidelity.

Specific QwenCloud APIs and Tools

Tool/Model	Purpose	Key Features	Integration Notes
Qwen 3.7 Max	High-level reasoning and text parsing	1M-token context window, text-only, high accuracy	Used for PDF distillation and scriptwriting



Tool/Model	Purpose	Key Features	Integration Notes
Qwen 3.7 Plus	Multimodal understanding and generation	Supports text, image, video; cost-effective	Used for scriptwriting, storyboarding, editing
Wan 2.7	Video generation and editing	3×3 grid synthesis, first/last frame control	Generates cinematic video from text and images
HappyHorse 1.0	Image-to-video and text-to-video	Realistic dynamic rendering, high detail	Produces fluid, natural videos
LangGraph	Human-in-the-loop workflows	Pause/resume agent execution, approval routing	Ensures human oversight at critical stages

Step-by-Step Workflow

- 1. PDF Upload and Parsing:** User uploads a PDF book; the parsing agent extracts and distills narrative elements, context, and structure.
- 2. Script Generation:** The scriptwriting agent generates a video script from the distilled content, ensuring narrative coherence and fidelity to the source.
- 3. Storyboard Creation:** The storyboarding agent translates the script into a visual storyboard, selecting scenes and framing shots.
- 4. Video Generation:** The video generation agent uses Wan and HappyHorse to create video clips from the storyboard, ensuring cinematic quality and narrative flow.
- 5. Video Editing:** The editing agent assembles clips, adds transitions, audio, and effects to produce the final video.
- 6. Human-in-the-Loop Review:** At each critical stage, HITL checkpoints allow human reviewers to pause, review, and refine the AI outputs.
- 7. User Interface Interaction:** Users browse the virtual bookshelf, select a book, and view the PDF and video in sync. In director mode, users add comments and prompts to refine the video.

Innovation Highlights

- **Autonomous End-to-End Pipeline:** The system autonomously handles the entire short drama creation pipeline from PDF to video, minimizing human intervention while ensuring quality.
- **Narrative Coherence and Complexity Handling:** The multi-agent system manages complex narratives with multiple characters and subplots by hierarchical story generation and prompt chaining, ensuring coherent and engaging video outputs.



- **User Feedback Integration:** Director mode allows users to provide targeted prompts and comments, which the AI incorporates to refine outputs, creating a collaborative human-AI workflow.
- **Efficient Token and Resource Management:** The architecture optimizes token usage by decomposing tasks and using cheaper Qwen 3.7 Plus for everyday workloads, reserving Qwen 3.7 Max for high-stakes reasoning tasks.

Real-World Impact and Scalability

- **Accessibility for Non-Readers:** Converts written content into engaging visual media, making stories accessible to users who prefer video over text.
- **Content Repurposing for Creators:** Enables creators to repurpose books, manga, and written stories into video formats for broader distribution and engagement.
- **Scalability:** The modular, multi-agent architecture can scale to handle large volumes of content, supporting open-source adoption and commercial use cases.
- **Ethical and Legal Compliance:** Incorporates HITL checkpoints to ensure content quality, prevent misuse, and comply with emerging regulations on AI-generated media.

Presentation and Documentation Plan

- **Demo Structure:** A live walkthrough of the interface showing the virtual bookshelf, PDF/video synchronized workspace, and director mode editing features.
- **Side-by-Side Comparison:** Demonstrates the PDF content alongside the generated video, highlighting narrative fidelity and visual quality.
- **Technical Documentation:** Includes architecture diagrams, code snippets, API usage examples, and a detailed explanation of the multi-agent collaboration and HITL integration.
- **Video Demonstration:** A 3-minute video showcasing the entire workflow from PDF upload to final video output, emphasizing the AI's narrative and multimodal orchestration skills.

Competitive Analysis

Feature	My Project	Competitor A (e.g., Sora)	Competitor B (e.g., Runway)
End-to-End Automation	Fully automated PDF-to-video pipeline	Focuses on text-to-video generation	Focuses on image-to-video and editing
		Limited to text prompts	Limited to image inputs



Feature	My Project	Competitor A (e.g., Sora)	Competitor B (e.g., Runway)
PDF-Specific Adaptations	Specialized PDF parsing and narrative extraction		
Collaborative Multi-Agent System	Uses multiple Qwen agents for script, storyboard, video generation, and editing	Single model or limited agent collaboration	Limited agent collaboration
Human-in-the-Loop Checkpoints	Integrated HITL for quality control and feedback	Limited or no HITL support	Limited HITL support
User Interface	Apple Books-style virtual bookshelf and workspace	Minimal or no UI integration	Minimal UI integration
Narrative Coherence	Handles complex narratives with hierarchical generation	Basic script generation	Basic script and storyboard generation
Video Quality and Editing	High-quality video generation with editing	High-quality video generation	High-quality video generation
Scalability and Cost Efficiency	Optimized token usage and cost-effective models	High cost and token usage	Moderate cost and token usage

Development Timeline

Phase	Tasks	Tools/Technologies	Estimated Time (Weeks)
PDF Parsing Agent	Develop PDF parser, narrative distiller, and context extractor	Qwen 3.7 Max, Alibaba Cloud	4
Scriptwriting Agent	Script generation from distilled content, narrative coherence checks	Qwen 3.7 Plus, LangGraph	3
Storyboarding and Scene Agent	Storyboard creation, scene selection, and visual planning	Qwen 3.7 Plus, Wan 2.7, HappyHorse	4
Video Generation Agent	Video clip generation, editing, and final assembly	Wan 2.7, HappyHorse, Qwen 3.7 Plus	5
Human-in-the-Loop Integration	Implement HITL checkpoints for script, storyboard, and video review	LangGraph, Microsoft Agent Framework	3



Phase	Tasks	Tools/Technologies	Estimated Time (Weeks)
User Interface Implementation	Develop Apple Books-style interface with virtual bookshelf and synchronized workspace	React, Tailwind CSS, Alibaba Cloud	
Testing and Refinement	End-to-end testing, user feedback incorporation, and quality assurance	Alibaba Cloud, Qwen models	4
Documentation and Demo	Create technical documentation, architecture diagrams, and demo video	Documentation tools, video editing	2

This comprehensive project plan aligns with the judging criteria for Track 2: AI Showrunner, emphasizing technical depth, innovation, problem value, and presentation. The proposed architecture leverages Qwen Cloud’s advanced AI models and video generation tools to create a production-ready, multimodal AI system that autonomously adapts written content into engaging visual storytelling.

